Resonant conversion of dark-photon fields in a plasma with a noise-free continuum kinetic method: bounds on nonlinear suppression and the role of collisions

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Whether the resonant conversion of dark-photon dark matter into plasma oscillations in the early Universe is cut off by nonlinear plasma dynamics has been argued, on the basis of particle-in-cell (PIC) simulations, to invalidate the cosmological constraints derived from it [1]. We test the claim in a noise-free setting: a Hermite–Fourier Vlasov solver for the canonical Maxwell–Proca–Vlasov system with kinetic mixing, built on SPECTRAX, in one spatial and three velocity dimensions, nonrelativistic. The method uses a pump-frame velocity basis with exact remaps, a field-scaled closure integrated exactly by a Lawson factor, and Dougherty collisions. We document the numerical limit of the method, a truncation instability of the asymmetrically weighted Hermite basis, which bounds the certified time windows (1000, 892 and 791.5 in units of ω_{pe}^{-1} at $v_q/v_{te} = 0.03$, 0.05, 0.07), and the alternative bases we tested and rejected. Inside these windows, at the strongest drive $v_q/v_{te} = 0.07$ the energy transfer follows the linear resonant law to 0.68 % with no resolved frequency detuning (bound $6.5 \times 10^{-6}\ \omega_{pe}$); a finite dark reservoir and a finite coherence time of the drive also leave the linear prediction unchanged to 0.81 % of the initial reservoir energy and 1.7 % of the ensemble work, respectively. The small residual is the back-reaction of the imposed seed fluctuations and scales with the square of their amplitude. With collisions, the kinetic absorption equals the linear Drude law with $\nu_{\rm eff} = (\nu_{ei}/2)(1 + m_e/m_i)$: the late rate is within 0.4 % of the steady-state power at $\nu_{ei} = 3\times10^{-2}\omega_{pe}$, and the work scales as $(v_q/v_{te})^p$ with $p = 1.99\pm0.10$. These are bounds, not a resolution of the cosmological question: they hold for $v_q/v_{te} \le 0.07$, $\omega_{pe}t \lesssim 790$ at the strongest drive, in a nonrelativistic one-dimensional model.

I. INTRODUCTION

A dark photon with kinetic mixing χ with the ordinary photon and a mass $m_{A'}$ is a well-motivated candidate for dark matter [2]. In the early Universe the dark-photon field oscillates coherently, and when $m_{A'}$ equals the plasma frequency of the ionized medium the field resonantly converts into longitudinal plasma oscillations, depositing energy in the plasma [3–5]. Cosmological constraints on dark-photon dark matter have been derived from the resonant transfer of a substantial part of the dark-photon energy density into the plasma, which heats it.

Hook, Huang and Shalaby (HHS) argued that the resonant transfer saturates because of plasma nonlinearities [1]. In their account, large-amplitude $k \simeq 0$ Langmuir waves, driven by the dark photon, excite higher-k Langmuir and ion-acoustic waves through the ponderomotive force; the resulting density inhomogeneities detune the resonance and limit the deposited energy to roughly the electron thermal energy, which would weaken the constraints by several orders of magnitude. Their evidence comes from relativistic one-dimensional PIC simulations of an electron–ion plasma with a prescribed uniform dark-photon drive.

A PIC calculation carries sampling noise, which seeds the very instabilities that the argument relies on. This paper asks a narrow question and answers it with a different numerical method: in a model with the same physical ingredients, when does the energy transfer from a resonant uniform drive depart from the linear resonant law, and how large is the departure that can be resolved? We use a noise-free continuum kinetic solver whose seeds are deterministic and whose work ledger closes to solver tolerance, so that any departure from the linear law can be attributed. We do not reproduce the HHS simulations: our model is nonrelativistic, the loading is deterministic, and the box and parameters differ in detail (Sec. VII).

The work consists of a methods part and a bounds part. The methods part (Secs. II–IV) describes the model, the Hermite–Fourier discretization, the verification suite, and the failure mode that limits the method. The results part addresses two research questions: (R1) does a finite dark-photon reservoir or a finite coherence time of the drive produce detuning or saturation of the energy transfer at the strongest drives we can certify (Sec. V); and (R2) with collisions present, does the sustained absorption obey the linear drag law, and what is missing before a cosmological statement (Sec. VI)? Failed and amended gates are reported where they affect a claim.

II. MODEL

A. Canonical Maxwell–Proca–Vlasov system

Each species s obeys the collisionless Vlasov equation in the combined force field,

$$\partial_t f_s + \mathbf{v}\cdot\nabla f_s + \frac{q_s}{m_s}\Big[\mathbf{E} + \eta\mathbf{E}_D + \mathbf{E}_{\rm drive} + \mathbf{v}\times(\mathbf{B} + \eta\mathbf{B}_D)\Big]\cdot\nabla_v f_s = 0, \quad (1)$$

with ordinary fields $\partial_t \mathbf{E} = c^2 \nabla \times \mathbf{B} - \mathbf{J}/\epsilon_0$, $\partial_t \mathbf{B} = -\nabla \times \mathbf{E}$ and canonical Proca fields

$$\partial_t \mathbf{E}_D = c^2 \nabla \times \mathbf{B}_D + \Omega_D^2 \mathbf{A}_D - \eta \mathbf{J}/\epsilon_0, \tag{2}$$

$$\partial_t \mathbf{B}_D = -\nabla \times \mathbf{E}_D, \quad \partial_t \mathbf{A}_D = -\mathbf{E}_D - \nabla \phi_D, \tag{3}$$

$$\partial_t \phi_D = -c^2 \nabla \cdot \mathbf{A}_D, \tag{4}$$

with constraints $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ and $\nabla \cdot \mathbf{E}_D + \Omega_D^2 \phi_D / c^2 = \eta \rho / \epsilon_0$. The ordinary current $\mathbf{J}$ sources both laws; the dark field acts back on the particles with strength η. Three modes are used: *ordinary* (no dark coupling), *prescribed drive* (a uniform $\mathbf{E}_{\rm drive} = \mathbf{E}_0 \cos(\omega t + \varphi)$ and an external-work ledger, no dark reservoir; the setting of HHS), and *self-consistent* (evolved Proca fields with a finite energy reservoir).

B. Mixing map

For a kinetic mixing $-\frac{\chi}{2} F_{\mu\nu} F_D^{\mu\nu}$ and a mass term on the dark field only, removing the mixing by a non-orthogonal shear gives the canonical coupling

$$\eta = \frac{\chi}{\sqrt{1-\chi^2}}, \qquad m_D = \frac{m_0}{\sqrt{1-\chi^2}}, \tag{5}$$

where m_0 is the bare mass and Ω_D is the dark-photon mass m_D expressed as a frequency. We checked this map against Sec. 1.1 of Ref. [2] (their $\varepsilon \equiv \chi$): the dark photon couples to the ordinary current with strength $e\varepsilon/\sqrt{1-\varepsilon^2}$. The sign convention differs by $\mathbf{A}_D \to -\mathbf{A}_D$, under which the model is symmetric in η. The model is not the massless "millicharged" choice of the same reference.

C. Energy and work ledger

In the parent normalization velocities are in units of c, $\Omega_0 \equiv \Omega_{c,0}$ and the species mass is $m_s = \Omega_0/\Omega_{cs}$, which is required for the energy theorem. With $\langle \cdot \rangle$ the box average the field energies and powers are

$$U_\gamma = \tfrac{1}{2} \Omega_0^2 \langle E^2 + B^2 \rangle, \tag{6}$$

$$U_D = \tfrac{1}{2} \Omega_0^2 \big\langle E_D^2 + B_D^2 + \Omega_D^2 (A_D^2 + \phi_D^2) \big\rangle, \tag{7}$$

$$P_\gamma = \Omega_0 \langle \mathbf{J} \cdot \mathbf{E} \rangle, \quad P_D = \Omega_0 \eta \langle \mathbf{J} \cdot \mathbf{E}_D \rangle, \tag{8}$$

$$P_{\rm ext} = \Omega_0 \bar{\mathbf{J}} \cdot \mathbf{E}_{\rm drive}. \tag{9}$$

The works W_γ, W_D, $W_{\rm ext}$ are integrated as extra ODE states with the same Runge–Kutta stages as the fields, so that $K - K_0 = W_\gamma + W_D + W_{\rm ext}$, $U_\gamma - U_{\gamma 0} = -W_\gamma$ and $U_D - U_{D0} = -W_D$ hold to solver tolerance. The Proca energy flux is $\epsilon_0 c^2 \mathbf{E}_D \times \mathbf{B}_D + \epsilon_0 \Omega_D^2 \phi_D \mathbf{A}_D$; omitting the $\phi_D \mathbf{A}_D$ term breaks the flux budget (Sec. IV). For a homogeneous plasma under the uniform drive the mean field obeys a driven oscillator at the plasma frequency $\omega_L = \sqrt{1 + m_e/m_i}\, \omega_{pe}$, whose solution is the *linear resonant law*: $\dot{\bar{E}} = -\bar{J}$, $\dot{\bar{J}} = \omega_L^2 (\bar{E} + E_0 \cos \omega t)$, $\dot{W} = \bar{J} E_0 \cos \omega t$. This law is exact in the homogeneous discrete Hermite system because only Hermite orders 0–1 enter (case H00, agreement to 10^{-9} for $\omega_{pe} t \le 1000$ even at quiver speeds of $15\, v_{te}$). With fluctuations present it is the baseline against which the transfer is measured, and the departure from it contains the $\langle \delta n\, \delta E \rangle$ back-reaction term of the mean-field identity.

D. Units and the HHS-like set-up

All simulations use $c = \omega_{pe} = n = 1$ for the electrons, $m_i/m_e = 1836$, $T_i = T_e$, $v_{te}^2 = 10^{-3} c^2$, box length $L = 40\, c/\omega_{pe}$ and a prescribed drive at $\omega = \omega_L$ unless stated. We call $v_q/v_{te} \equiv E_0/(v_{te}\omega_{pe})$ the effective initial force (the quiver speed grows secularly with time at resonance). Energies are quoted in units of nT_eV. HHS quote $v_q^D/v_{te} = 0.03$ and 10^{-3} for their resonant runs, $m_i/m_e = 1836$, $T_e = T_i = 10^{-3} m_e c^2$, $L = 40\, c/\omega_{pe}$ and 1000 cells (v1 text, Appendix B); we use the same v_{te}, mass ratio, temperature ratio and box length, but not the loading or the relativistic pusher. Runs here are *HHS-inspired*, not reproductions.

III. METHOD

A. Hermite–Fourier discretization

The velocity dependence is expanded in Cartesian asymmetrically weighted (AW) Hermite functions with independent orders (N_n, N_m, N_p), widths α_s and centres u_s; space is Fourier, with the strict 2/3 dealiasing mask of the parent (which retains $3|m| < N_x$, so every grid size is alias-free for quadratic products). A single call to the SPECTRAX Hermite–Fourier kernel per right-hand-side evaluation receives the physical force fields $\mathbf{E} + \eta \mathbf{E}_D + \mathbf{E}_{\rm drive}$ and $\mathbf{B} + \eta \mathbf{B}_D$, and the current is evaluated once for both Ampère laws. Time integration is Dopri8 with a PID controller (Diffrax). The kinetic-energy ledger closes only when every forced direction has Hermite order at least 3. The parent is SPECTRAX [6]; the three studies reported here were run at the parent commits recorded in each record (R1: `3de6b2b`, R1 nonlinear: `4090fbe`, R2: `35231ab`), all on the integration branch `dark-baseline`, which is a set of open parent PRs, not a release.

B. Pump frame and basis remaps

With `frame="pump"` the state carries the centres u_s and widths α_s. Each centre follows the total uniform force, $\dot{u}_s = (q/m)_s (\bar{\mathbf{E}} + u_s \times \bar{\mathbf{B}})$, with the box averages including $\eta \mathbf{E}_D$ and the drive; the parent kernel then uses $\mathbf{E} - \bar{\mathbf{E}}$. This is an exact change of variables: the coherent $k = 0$ quiver sits in $u_s(t)$ rather than in high Hermite orders. Widths and residual drifts change only between

integration segments, through the parent's exact lower-triangular remap (every moment below the truncation is preserved); triggers fire on $|U-u|/a > 0.3$ or $|a'/a-1| > 0.05$, with caps $|\Delta u| \le a'$, $a/a' \le 1.1$, $a' \ge 1.1\sigma$.

C. Closures

Truncation needs a closure. The parent closure is a hypercollision $\propto n(n-1)(n-2)$ that leaves Hermite orders 0–2 unchanged. We use the field-scaled form of SPECTRAX PR #66,

$$\nu_s = c \left|\frac{q}{m}\right|_s \frac{\sqrt{2N_n}\,\max|E-\langle E\rangle|}{a_s} \times \frac{n(n-1)(n-2)}{(N_n-1)(N_n-2)(N_n-3)}, \quad (10)$$

with strength c (density, momentum and energy rows untouched; for magnetized runs a Lorentz-force-scaled variant, PR #72, is used, not needed here). For $c = 64$ the closure rate is the stiff eigenvalue of the system: the Jacobian's spectral radius equals the closure rate ν_e (real and negative) to the printed precision, so the explicit step is bounded by $5.17/\nu_e$. We therefore integrate the diagonal closure damping exactly with a Lawson integrating factor wrapped around Dopri8 (SPECTRAX PR #81; the original Lawson reference is **[TODO: verify and cite]**, exponential integrators for kinetic equations are reviewed in Ref. [7]): with $w = e^{r(t-t_0)}y$ and the rate r frozen at the step start the split is an identity, the order is kept, and only the variation of ν over the step stays explicit. For $c = 64$ this reduces the step count by a factor 5.0 to $t = 500$ with agreement to 5×10^{-10}; for $c = 4$, where the closure is not stiff, it gives 0.96 (no gain). An exact Hermite-streaming factor was also built (PR #83) and rejected: it gave no step reduction and cost 2.7 times more per step. Table II lists these and other negative results.

D. Collisions

Collisions use the conservative Dougherty (Lenard–Bernstein–Fokker–Planck) operator of Hakim et al. [8] with the multispecies forms of Francisquez et al. [9]. We use two cross-species conventions: LBO-EM (parent `dougherty`, the form of Francisquez et al. Eqs. 2.18–2.20, momentum rate equal to the Morse rate and a temperature-difference rate equal to twice the Morse rate) and LBO-G with $\beta = 0$ (parent PR #73, both rates equal to the Morse rates). The collision frequencies are velocity independent, $\nu_{ee} = \nu_{ei}$, $\nu_{ii} = \nu_{ei}\sqrt{m_e/m_i}$, $m_e\nu_{ei} = m_i\nu_{ie}$. Collisions do no net work and conserve mass, momentum and energy in the homogeneous controls to round-off.

IV. VERIFICATION AND LIMITS

A. Verification

Table I and Fig. 1 summarize the checks. The linear Landau roots of the Maxwell–Proca–Vlasov system (the B00 case, $k\lambda_{De} = 0.2$–0.7, $\eta = 0.3$, $\Omega_D = \omega_{pe}$) agree with SciPy Faddeeva roots of $D_L = (Q - \Omega_D^2)(1+\chi) + \eta^2 Q\chi$ to 1.3×10^{-4} across 32 runs, but at $k\lambda_{De} = 0.2$ the damping rate ($\sim 5 \times 10^{-5}$) is not resolved inside the Hermite-limited fit window; only the frequency is validated there. The oscillating two-stream instability that sets the high-k cascade is recovered from the Hermite Floquet analysis against the exact kinetic dipole-pump relation (Silin; Nishikawa; **[TODO: verify and cite the original references]**) at 22 unstable points to 1.3 %; with fixed ions every rate vanishes, so the instability needs mobile ions. This is known physics in a new regime (resonant pump with a secular excursion and $T_i = T_e$). Further checks, all in the repository: cold coupled oscillators to 10^{-11}, Proca vacuum and Yukawa limits, the η-sign symmetry, the exact mean pump, a Weibel and filamentation set against an independent kinetic dispersion relation including Proca coupling (within 10^{-3}), magnetized rotation and cold electromagnetic eigenmodes (within 10^{-3}), and Dougherty collisional relaxation against the exact rates (to 10^{-13}). Independent discretizations: a NumPy semi-Lagrangian grid solver (and its Vlasov–Ampere–Proca extension) agrees with the Hermite echo amplitude to 2×10^{-5}; the Dark-JAX-in-Cell PIC code [10] approaches the Hermite Landau fit as cells and particles increase, with a remaining difference of order 1 % in damping set by PIC noise (Fig. 1c); this is not a claim of PIC convergence. GPU and CPU agree to 9.7×10^{-9} on one case only. Gradients of converged dark observables are verified (Table I); reverse mode through Diffrax's `RecursiveCheckpointAdjoint` was found wrong (2–15 %) for many constant-step saves, and `DirectAdjoint` is used.

B. The Hermite truncation instability and the certified windows

The method has one limit that matters for the rest of the paper. In a non-uniform field the truncated AW Galerkin operator $-v\partial_x + (q/m)E\partial_v$ is not anti-self-adjoint; its frozen-field spectrum has growing modes in the top third of the Hermite ladder, with a rate that rises with N_n (Fig. 2a; at $K = 5$, $N_n = 128$, $E_1 = 10^{-2}$ the largest growth rate is 1.6, against $\le 8 \times 10^{-15}$ for the symmetric basis, which is neutral to round-off). This is a numerical instability of the basis, not physics. It is distinct from the physical oscillating two-stream instability, which the Hermite model reproduces and which amplifies finite-k fields strongly at resonance (the Hermite model understates the growth: Nn 32/64/128 agree with each other and end 1.3–4 times below the exact solution). The

TABLE I. Verification summary (records: `docs/results.md` and the cited JSON files). The ladder, including failed gates, is in `docs/status.md`.

Check	Result (record)
Linear Landau roots, ordinary and dark, $k\lambda_{De} = 0.2$–0.7	worst relative error 1.3×10^{-4} vs. SciPy Faddeeva roots; ledger defect $\leq 3 \times 10^{-18}$
OTSI growth vs. Silin–Nishikawa kinetic relation	22 unstable points, worst difference 1.3%
Dark-JAX-in-Cell PIC, Landau $k\lambda_{De} = 0.5$	rms $\Delta \ln \lvert E_k \rvert$ falls 0.22→0.13 from 32 to 128 cells; PIC noise limits agreement to ∼1%
GPU vs. CPU, one case ($T = 100$)	max relative difference 9.7×10^{-9} (adaptive trajectories)
AD vs. finite differences	reverse = forward to 2×10^{-13}, analytic to 5×10^{-10} (`studies/c09/run.json`, 19 derivatives)

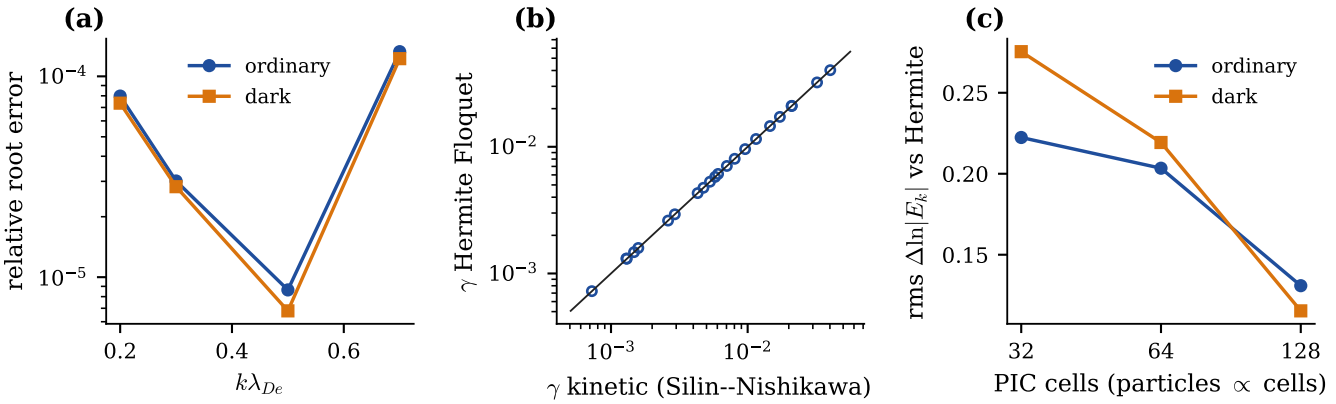


FIG. 1. Verification. (a) Relative error of the fitted complex frequency against the independent kinetic root, ordinary and dark ($\eta = 0.3$), $N_n = 128$, $N_x = 8$ (`docs/_static/b00/run.json`). (b) Hermite Floquet growth rate against the exact kinetic Silin–Nishikawa relation for the oscillating two-stream instability (`studies/instab/otsi.json`). (c) RMS difference of $\ln \lvert E_k \rvert$ between Dark-JAX-in-Cell and Hermite on $2 \leq \omega_{pe} t \leq 12$ for Landau damping, for increasing cells and particles (`studies/c05/run.json`).

AW instability is excited by the large-amplitude pump and ends the simulation by loss of positivity or by the step budget.

Certified windows are defined by gates declared before the runs: refinement pairs (N_x, N_n, closure strength) must agree to 10 % in ΔK_e and W_{ext}, the electron distribution must stay positive ($K_s > 0$ and $k = 0\, T_x > 0$), the solver must reach the time, and the energy ledger must close. The window is the minimum of these times. It is therefore a lower bound on where the Hermite results can be trusted, not a statement about physics.

Alternatives were tested and rejected, all under predeclared gates (Table II). Higher resolution on GPU ($N_x \in \{64, 128\}$, $N_n \in \{128, 256\}$, $v_q/v_{te} = 0.1$ and 0.05) failed the extension gate G1 and ended every run inside the instability at 328–595 ($v_q/v_{te} = 0.1$) and 440–751.5 ($v_q/v_{te} = 0.05$): the pairs agree to the end of the shorter run, so this is a failure of the resolution route rather than a disagreement. In a frozen field no closure strength up to 1024 brings the AW growth below 10^{-3} at peak field 4×10^{-3} (39/39 points fail), leaving a floor that grows as $K^{0.15} N_n^{0.30} E^{1.26}$, so a resolution-consistent closure strength does not exist for this closure family. A stable Galerkin form of Dai [11] is unstable in the AW coordinates in float64 at $N_n \geq 64$ (the AW-to-orthonormal change of basis has condition $\sim 2^N$) but stable and Silin-accurate in symmetric Hermite functions of width $a/\sqrt{2}$; the symmetric-weight basis itself, and the anti-symmetric square-root form of Issan et al. [12, 13], are linearly neutral to round-off and Silin-accurate, but were rejected under their declared nonlinear $v_q/v_{te} = 0.1$ gates (symmetric weight: every run blows up at $t = 637$–738 and the refinement pair agrees only to 634; square-root form: departure from the grid reference at 391 and 498, earlier than the symmetric weight). Production therefore stays with the AW basis and the closure of PR #66, and the windows below are its windows.

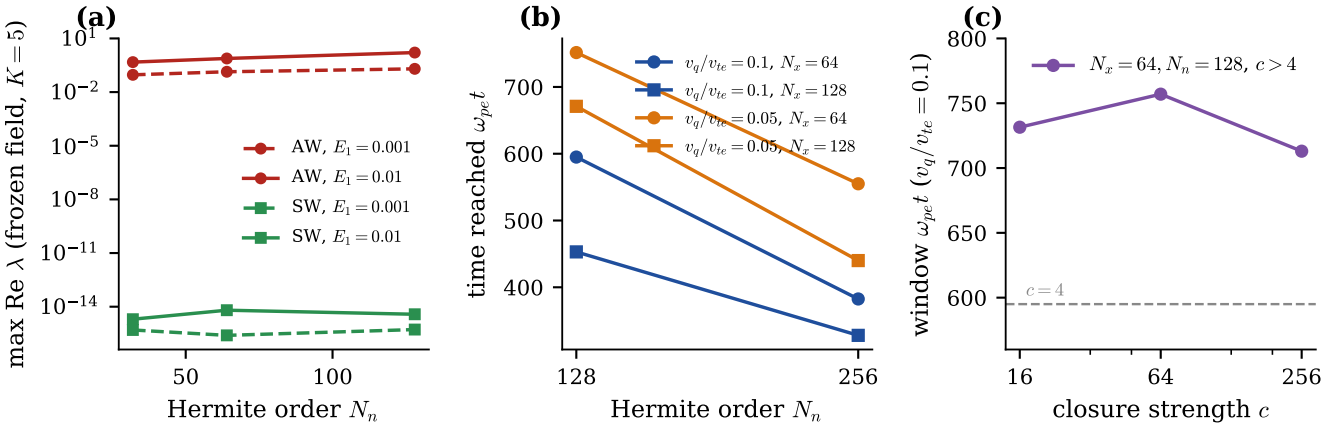


FIG. 2. The limit of the method. (a) Largest real part of the frozen-field spectrum for the AW basis (grows with N_n) and the symmetric-weight basis (round-off), $K = 5$ (`studies/highk/eig.json`). (b) Time reached by AW runs with closure $c = 4$ at $v_q/v_{te} = 0.1$ and 0.05 for $N_x = 64, 128$ and $N_n = 128, 256$, all ended at the step budget inside the instability: more resolution ends the run sooner (`studies/gpures/summary.json`). (c) At $v_q/v_{te} = 0.1$, $N_x = 64$, $N_n = 128$ a stronger fixed closure extends the window from 595 to a ceiling of 713–757 (`studies/rclosure/stagef.json`).

TABLE II. Negative and amended results that bound the claims. "Declared" means the gate was committed before the run and not amended.

Study	Outcome	Consequence
GPU resolution extension	G1/G4 failed at $v_q/v_{te} = 0.1$, 0.05; every run ends at the step budget	no new window; need a stable basis
Closure-strength route	Q1 failed; floor in frozen field	scaled-closure option not built
Stiffness vs instability	stiff eigenvalue is the closure; instability max Re $\lambda = 2$–3 is not	integrator is not the limit
Lawson (closure)	D2 failed as declared ($c = 4$: no gain)	useful for $c = 64$ only
Lawson (streaming)	S4 failed	rejected
Dai / symmetric weight / square-root	gates failed (Sec. IV)	AW + closure stays
R1 coherence v1	declared variance wrong (realized τ was 2× nominal)	v2 declared and used
R1 nonlinear F2	hypothesis "homogeneous truncation effect" rejected	deficit is seed back-reaction
R2 theory check T0	window longer than the record	T0b (follow-up) used
R2 K2	30/36; six failures at $\nu_{ei} = 10^{-3}$ detuned	seed effect, F1 passed
H08 G1, G4	restart chunk length; T_i responds to halved exchange	recorded, not amended
B11 G0, G3, G3b'	mis-specified gates; round-off	recorded; ledger closes to 10^{-17}

V. R1: FINITE RESERVOIR AND FINITE COHERENCE

A. Design

Two idealizations of the prescribed-drive calculation are removed. First, in the self-consistent mode the dark photon is a free $k = 0$ Proca mode with $\Omega_D = \omega_L$ whose energy reservoir U_{D0} is finite and is depleted by the transfer; we use $u = U_{D0}/(nT_eV) = 0.1, 1, 10$ with η chosen so that the initial effective force matches the prescribed drive. Second, the dark field has a finite coherence time: the prescribed drive is $E_0 \cos(\omega_L t + \Phi(t))$ with Φ a phase random walk of independent Gaussian frequency samples, giving a Lorentzian line of half-width $1/\tau_c$ and no phase resets. The comparison is always with the exact linear law for the same drive (and, for the reservoir, the linear coupled oscillator $(\bar{E}, \bar{J}, E_D, A_D)$, whose matrix exponential agrees with an ODE solver to 6.6×10^{-11}).

B. Certified windows

Table III gives the windows: 1000, 892 and 791.5 for $v_q/v_{te} = 0.03$, 0.05 and 0.07. The windows are set by positivity loss of the $N_x = 64$ members, that is, by the AW truncation instability reaching the resolved range, not by disagreement of the refinement pairs. The strongest certified drive is $v_q/v_{te} = 0.07$ with window 791.5. Drives above that, including $v_q/v_{te} = 0.1$, have no window beyond about 760 (Sec. IV).

C. Linear transfer and the seed back-reaction

At $v_q/v_{te} = 0.07$ the work $W_{\rm ext}$ follows the exact linear resonant law over the whole window (Fig. 3a): the maximum of $|W_{\rm ext}/W_{\rm lin} - 1|$ over four resolutions is 0.68 %, the four members agree on it to 2×10^{-4}, and at the window end $W_{\rm ext} = 381.3$ against $W_{\rm lin} = 384.0$ (in nT_eV). The mean-field frequency shift, measured as the slope of the lock-in phase difference, is at most $6.5\times10^{-6}\,\omega_{pe}$ (the declared resolution threshold was $10^{-3}\omega_L$; not resolved). No saturation and no detuning are resolved. The departure rule (a 10 % departure in the coarsest member confirmed at 5 % in all refinements) is not met.

The residual deficit is not a drive effect. At the three drives it is the same to three digits (it grows as $-10^{-8}\,t^2$) until the oscillating two-stream stage, ending at -1.07, -0.88 and -0.68 % at the window ends. A seed-scale test at $v_q/v_{te} = 0.07$ (the initial density fluctuations multiplied by s) shows the deficit scales as s^2 (Table IV, Fig. 3b): the ratio of $s = 0.1$ to $s = 1$ is 0.0100 and the unseeded run gives 2.3×10^{-9} at $\omega_{pe}t = 400$. The deficit is therefore the back-reaction of the quiver-advected seed modulation on the mean field (the $\langle\delta n\,\delta E\rangle$ term of the mean-field identity), set by the imposed fluctuation level, not by the pump. A physical fluctuation level different from 5×10^{-4} would rescale it. Our earlier hypothesis that it is a truncation effect of the seedless homogeneous system was tested (follow-up F2) and rejected: the unseeded deficit is only 1.4×10^{-6} of the seeded one. Only the seed scales 0, 0.1 and 1 at $t \le 400$ were run.

D. Energetics and the onset of the oscillating two-stream instability

Why the transfer stays linear is energetic. At the window end of $v_q/v_{te} = 0.07$, 371 of the $W_{\rm ext} = 381.3\,nT_eV$ resides in coherent $k = 0$ electron quiver kinetic energy; the thermal gain of the electrons is $0.24\,nT_eV$ and the ions receive 0.05 % of $W_{\rm ext}$. The fluctuating-field amplitude ($\sum_{k\ge1}|E_k|^2$) reaches about $2077\,E_0^2$ only when the oscillating two-stream instability turns on, at $\omega_{pe}t = 783.5$ ($N_x = 32$) to 862 (higher N_x and N_n members; onset defined in the gates, compared within equal N_x). After that the run ends by positivity loss. Within the window the instability has not extracted a resolvable fraction of the work. *This is the main limitation of the bound*: it says that no saturation appears *before* the cascade of fluctuations starts, and nothing about what happens after.

E. Finite reservoir

In the self-consistent mode (Table V) the dark energy $U_D(t)$ follows the linear coupled-oscillator prediction at all three drives: within $0.039\,U_{D0}$ (maximum over six cases, $u = 0.1, 1, 10$) at $v_q/v_{te} = 0.01$ and 0.03 over the windows 1000 and 940 (`studies/r1/reservoir_summary.json`), and within 0.23 % and 0.81 % of U_{D0} at $v_q/v_{te} = 0.07$ for $u = 1$ and 10 over the window 791.5. The beat times equal π/η ($\eta = 0.049$ and 0.016 for $u = 1, 10$). The accumulated work W_D follows the matched prescribed drive until the split time $t_{\rm split} = 50.5$ and 72 (equal to the linear values): the reservoir gives up 47 % of its energy for $u = 1$ and exchanges energy through beats for $u = 10$, with W_D within 1.0 % and 2.7 % of the linear prediction while it is above 20 % of its running maximum. Finite reservoir depletion is therefore captured by the linear theory; no nonlinear reservoir response is resolved. The ledger closes to 1.7×10^{-9} over all 33 stage-1 and stage-2 runs (gate 10^{-6}).

F. Finite coherence

Figure 3c shows the ensemble ratio of the kinetic to the linear same-drive work. At $v_q/v_{te} = 0.01$ (window 1000, v2 definition with τ_c the field coherence time) the ratios are 1.000 ± 0.007, 1.089 ± 0.064 and 0.998 ± 0.003 for $\tau_c\omega_{pe} = 30, 100, 300$; the value at $\tau_c\omega_{pe} = 100$ is 1.4 standard errors above unity and is consistent with the sampling spread of six realizations. At $v_q/v_{te} = 0.07$ the ratios are 1.0050 ± 0.0058 and 1.0000 ± 0.0038 for $\tau_c\omega_{pe} = 100$, 300, with bounds on the ensemble nonlinear correction (including twice the standard error) of 1.7 % and 0.8 %; individual realizations deviate more (up to 4.7 % and 1.5 % of the peak linear work), with ratios on both sides of unity. The late work rate saturates at about $E_0^2\tau_c\omega_L^2/4$ instead of growing, as in the linear theory. The first (v1) coherence runs realized a phase variance dt/τ_c rather than the declared $2\,dt/\tau_c$, i.e. a coherence time twice the nominal; the records were kept and relabelled and a corrected v2 was declared and run (Table II).

G. Scope of R1

The bounds hold for one box ($L = 40$), one ion mass, $T_i = T_e$, drives up to $v_q/v_{te} = 0.07$, windows set by the AW instability, fluctuation seeds of one amplitude (5×10^{-4}), a Lorentzian line only, and reservoir and coherence

TABLE III. Certified windows for the strong-drive set-up with closure $c = 64$ integrated exactly (`studies/r1nl/stage1_summary.json`). Members: $(N_x, N_n) = (32, 64), (64, 64), (32, 128), (64, 128)$ and $(32, 64)$ at $c = 32$. The window is the minimum of refinement-pair times, positivity times, times reached and the cap $T = 1000$; the grid column is the masked spectral-grid reference (reported, not part of the window).

v_q/v_{te}	window $\omega_{pe}t$	limiter	max ledger (B)	grid agrees with B until
0.03	1000	cap $T = 1000$	6.6×10^{-12}	1000 / 1000
0.05	892	positivity, $N_x = 64$	3.3×10^{-12}	979.5 / 962.5
0.07	791.5	positivity, $N_x = 64$	9.7×10^{-13}	870 / 863

TABLE IV. Seed-scale test of the deficit $D = W_{\rm ext}/W_{\rm lin} - 1$ at $v_q/v_{te} = 0.07$, $N_x = 32$, $N_n = 64$ (`studies/r1nl/followup_summary.json`); the seed amplitudes are multiplied by s.

seed scale s	$D(100)$	$D(200)$	$D(300)$	$D(400)$
1	-1.03×10^{-4}	-4.15×10^{-4}	-9.35×10^{-4}	-1.66×10^{-3}
0.1	-1.03×10^{-6}	-4.15×10^{-6}	-9.36×10^{-6}	-1.66×10^{-5}
0 (unseeded)	1.00×10^{-10}	8.58×10^{-10}	1.58×10^{-9}	2.27×10^{-9}

not combined in one run. Not done: a Gaussian line, reservoir and coherence combined in one run, windows beyond about 790 at 0.07, $v_q/v_{te} > 0.07$, seed levels other than 5×10^{-4} (only scales 0, 0.1 and 1 at $t = 400$), and inhomogeneous dark fields. The grid reference (masked spectral grid) agrees with the Hermite coarse member until 870 and 863 at $v_q/v_{te} = 0.07$ (Table III).

VI. R2: COLLISIONS

A. Linear drag law

With collisions the uniform relative velocity relaxes at the Morse rate, identical in LBO-EM and LBO-G ($\beta = 0$),

$$\nu_{\rm eff} = \frac{\nu_{ei}}{2}\Big(1 + \frac{m_e}{m_i}\Big), \tag{11}$$

so the uniform driven cold response obeys $\dot J = \omega_L^2(\bar E + E_D) - \nu_{\rm eff} J$ and the steady absorbed power per volume is

$$P_{\rm ss} = \frac{\omega_L^2\, \nu_{\rm eff}\, \omega^2 E_0^2}{2\,[(\omega_L^2 - \omega^2)^2 + \nu_{\rm eff}^2 \omega^2]}, \tag{12}$$

$$P_{\rm ss}^{\rm res} = \frac{\omega_L^2 E_0^2}{2\nu_{\rm eff}}, \tag{13}$$

with the transient decaying as $\exp(-\nu_{\rm eff} t/2)$ (1/e time $4/\nu_{ei}$). This is the cold Drude response of Dubovsky and Hernández-Chifflet [3], who write the friction as $-m_e\nu(u_e - u_i)$ with a Spitzer-like ν and find a shielding suppression $\propto m^2/\omega_p^2$ off resonance; and it matches the non-resonant absorption rate of McDermott and Witte [4]. Their ν corresponds to $\nu_{\rm eff}$ here. In our model ν is *constant*: the Spitzer scaling $\nu \propto T_e^{-3/2}$ is absent.

B. Design and gates

The set-up is that of R1 (mobile ions, pump frame, $c = 64$ with Lawson, $T = 1000$) with Dougherty e–e, e–i and i–i collisions in both conventions. The grid is $v_q/v_{te} = 0.01, 0.03, 0.05$ by $\nu_{ei} = 10^{-3}, 3\times10^{-3}, 10^{-2}, 3\times10^{-2}$ by $\omega = \omega_L, 0.9, 1.1$ (in ω_{pe}): 192 runs in all, including $N_x = 64$ and $N_n = 128$ refinement members and unseeded controls. Gates were committed before the runs. Every EM case reaches $t = 1000$ with refinement pairs agreeing to the end and a refinement spread of the late rate $\le 2.2 \times 10^{-4}$. The nonrelativistic window ($T_e/T_e(0) < 90$, i.e. $v_{te} < 0.3c$) is shorter only for $v_q/v_{te} = 0.05$ resonant. Rates are evaluated at $t_b = 996.5$.

C. Kinetic against linear

Table VI and Fig. 4 give the result. Where collisions act and the linear transient has decayed, the kinetic absorption equals the linear drag law: at $\nu_{ei} = 3\times10^{-2}$ the late rate is within 0.4 % of $P_{\rm ss}$ for all three frequencies and drives, and gate K3 (steady-state law) passes in 12 of 12 cases where $P_{\rm lin}$ is within 10 % of $P_{\rm ss}$. For the other cases the kinetic rate equals the exact *transient-inclusive* linear law to 5 % except in six cases, but the steady state is not reached within $t = 1000$: for $\nu_{ei} = 10^{-3}, 3\times10^{-3}, 10^{-2}$ the resonant rate is only 0.20, 0.48, 0.89 of $P_{\rm ss}$ (the amplitude transient has 1/e time $4/\nu_{ei}$). The sustained law is therefore tested directly at $\nu_{ei} = 3\times10^{-2}$ (and 10^{-2} detuned), and through the linear ODE for the rest. Gate K2 passes in 30 of 36 cases.

The six failures are the detuned $\nu_{ei} = 10^{-3}$ cases, where the kinetic late rate is 8–14 % high (mean +14 % at 0.9 and +9 % at 1.1), independent of v_q/v_{te}. In the unseeded controls the kinetic and linear laws agree to 3×10^{-8}, and a seed-scale follow-up (F, declared after

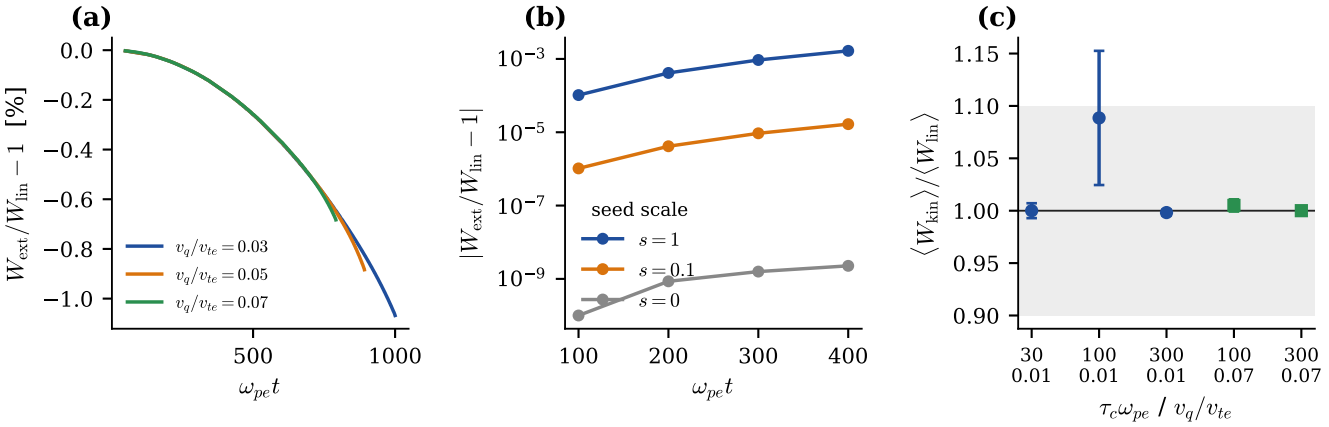


FIG. 3. R1. (a) $W_{ext}/W_{lin} - 1$ for the $N_x = 32$, $N_n = 64$ member at the three drives, each to its certified window; the three curves coincide until the oscillating two-stream stage. (b) The deficit at $v_q/v_{te} = 0.07$ for seed scales $s = 1$, 0.1, 0 (`studies/r1nl/followup_summary.json`). (c) Finite coherence: ensemble ratio $\langle W_{kin}\rangle/\langle W_{lin}\rangle$ with standard error over six realizations for $\tau_c\omega_{pe} = 30, 100, 300$ at $v_q/v_{te} = 0.01$ (window 1000) and $100, 300$ at $v_q/v_{te} = 0.07$ (window 791.5); band: $\pm 10\,\%$ (`studies/r1/coherence2_summary.json`, `studies/r1nl/stage2_summary.json`).

TABLE V. Finite reservoir (self-consistent dark field, $\Omega_D = \omega_L$), $v_q/v_{te} = 0.07$ (`studies/r1nl/stage2_summary.json`, N4).

$u = U_{D0}/nT_eV$	η	max $\lvert U_D - U_D^{lin}\rvert/U_{D0}$ [%]	t_{split}	W_D bound [%]
1	0.049	0.23	50.5	1.0
10	0.016	0.81	72	2.7

the main analysis and before its runs) shows the departure scales as s^2 (ratios 0.0105 and 0.0099 for $s = 0.1$, against $s^2 = 0.01$; F1 passed): it is the same seed back-reaction as in R1, large only where W_{ext} is itself tiny. The resonant late rate is low by 1.5–4 % at $\nu_{ei} = 10^{-3}$ (0.963 at the lowest) and by $\leq 0.2\,\%$ at 3×10^{-2}; at $v_q/v_{te} = 0.05$ resonant the deficit (3.7 %) is larger than at 0.01 and 0.03, but drive amplitude and heating are not separable in these data, so a few-percent heating or drive correction is allowed but not attributed.

D. Deposition law

Fitting $\ln P = a + p\ln(v_q/v_{te}) + q\ln\nu_{ei}$ over the 12 cases per frequency (standard errors from the residual scatter, so they include the transient-state curvature) gives for resonant, 0.9 and 1.1: $p = 1.99 \pm 0.10$, 2.00 ± 0.03, 2.00 ± 0.02 (kinetic), against 2 (linear, exact), and $q = -0.51 \pm 0.05$, 0.75 ± 0.02, 0.88 ± 0.01 (kinetic) against -0.52, 0.79, 0.90 (linear). The amplitude scaling is quadratic. The ν_{ei} scaling follows the transient-inclusive linear law, not a pure power: the detuned rate grows with ν (off-resonant absorption proportional to ν, until the beat transient decays), and the resonant rate falls with ν with an exponent near $-1/2$ because the steady state is not reached. The geometric-mean correction of kinetic to linear power is $\kappa = 0.985 \pm 0.003$, 1.040 ± 0.017, 1.025 ± 0.011. The domain of the law is $v_q/v_{te} \leq 0.05$, $\nu_{ei} \in [10^{-3}, 3\times10^{-2}]\omega_{pe}$, $\omega \in \{0.9, \omega_L, 1.1\}\omega_{pe}$, $t \leq 1000$, one box and mass ratio, one seed level.

E. Collision form and heating

EM and G ($\beta = 0$) agree in W_{ext} to 2.5×10^{-6} and the late-rate ratio differs by $\leq 7 \times 10^{-6}$ (agreement checks pass in 36 of 36 cases); electron kinetic energy and temperature differ by $\leq 8 \times 10^{-3}$. Ion temperature differs by up to 31 % and the electron–ion energy exchange is halved, as expected: the absorption is set by the momentum friction, identical in the two forms, while the exchange is the temperature-difference rate. The ion-heating gate of the earlier H08 pilot failed as declared for this reason (Table II). Electron heating is large: $T_e/T_e(0)$ reaches 231 (Fig. 4d), beyond $v_{te} = 0.3c$ for $T_e/T_e(0) > 90$, i.e. outside the nonrelativistic model. Heating does not change the rate in this model because ν is constant (R_{late} stays within 1 % of unity while $T_e/T_e(0)$ rises from 7 to 70 at $v_q/v_{te} = 0.03$, $\nu_{ei} = 10^{-2}$, resonant). Ion heating is small ($T_i/T_i(0) \leq 2.8$) and the $e \to i$ exchange is 4×10^{-5} to 1.6×10^{-2} of W_{ext}, growing in proportion to ν_{ei}. A declaration defect in the theory self-check T0 (averaging window longer than the record) was recorded; T0b, declared as a follow-up, passes with deviations from P_{ss} of 3×10^{-5}.

TABLE VI. Late-time absorbed power, kinetic ($N_x = 32$, $N_n = 64$, LBO-EM) against the exact transient-inclusive linear law, $v_q/v_{te} = 0.03$ (`studies/r2/summary.json`). $f_{ss} = P_{lin}/P_{ss}$; $T_e/T_e(0)$ at t_b; K2: kinetic equals linear within 5 %.

ω/ω_{pe}	ν_{ei}/ω_{pe}	P_{lin}/P_{ss}	P_{kin}/P_{ss}	P_{kin}/P_{lin}	$T_e/T_e(0)$	K2
res.	0.001	0.20	0.192	0.9830	35.2	pass
res.	0.003	0.48	0.472	0.9841	68.6	pass
res.	0.01	0.89	0.877	0.9908	83.6	pass
res.	0.03	1.00	0.996	0.9979	48.1	pass
0.9	0.001	2.00	2.279	1.1406	1.02	FAIL
0.9	0.003	1.54	1.584	1.0281	1.05	pass
0.9	0.01	1.11	1.106	0.9994	1.12	pass
0.9	0.03	1.00	1.003	1.0000	1.32	pass
1.1	0.001	1.38	1.496	1.0857	1.02	FAIL
1.1	0.003	1.21	1.224	1.0145	1.05	pass
1.1	0.01	1.04	1.037	0.9996	1.14	pass
1.1	0.03	1.00	1.000	1.0000	1.38	pass

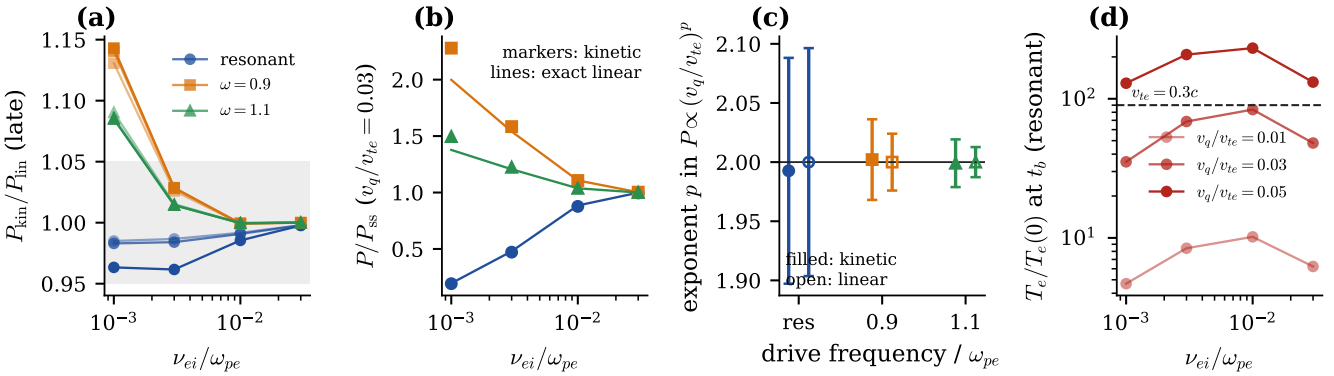


FIG. 4. R2. (a) Ratio of the kinetic to the exact linear late-time power for $v_q/v_{te} = 0.01, 0.03, 0.05$ (increasing opacity) and three drive frequencies; band: the 5 % gate. (b) Late power relative to the steady state P_{ss} at $v_q/v_{te} = 0.03$, kinetic (markers) and linear (lines). (c) Fitted exponent p of the drive amplitude, kinetic (filled) and linear (open). (d) Electron heating at t_b for the resonant cases; dashed: $v_{te} = 0.3c$, beyond which the nonrelativistic model is outside its validity. All from `studies/r2/summary.json`.

F. What is needed for a cosmological statement

Our collisions fix ν at a constant. A cosmological deposition law needs (i) the velocity-dependent Coulomb frequency $\nu(T_e) \propto T_e^{-3/2}$, which couples the heating to the absorption and lowers it; (ii) radiative and Compton cooling; (iii) cosmological expansion and ionization history; (iv) relativistic electrons, since $T_e/T_e(0)$ exceeds the nonrelativistic limit; (v) the $k \neq 0$ coupling of a real dark-photon field; and (vi) a steady-state measurement at $\nu_{ei} \leq 10^{-2}$ resonant, which needs $t \gg 4/\nu_{ei}$. ν_{ee} and ν_{ii} are order-of-magnitude choices that do not enter the uniform drag.

VII. DISCUSSION

What the results do establish. Within a one-dimensional, nonrelativistic, deterministic-seed model with the ingredients of HHS, and for the drives and times listed in Table III, the energy transfer from a uniform resonant drive follows the linear resonant law: $v_q/v_{te} = 0.03$ to $\omega_{pe}t = 1000$, 0.05 to 892 and 0.07 to 791.5. At the strongest drive the nonlinear correction to W_{ext} is bounded by 0.68 %, and the work has reached $W_{ext} = 381.3\, nT_eV$ (111, 247 and 381 nT_eV at the window ends for 0.03, 0.05, 0.07), almost all of it coherent $k = 0$ quiver energy. A finite reservoir and a finite coherence time of $\tau_c\omega_{pe} \geq 100$ do not change this. Collisions give the linear Drude law, with the kinetic rate equal to the linear law to a few per cent and to 0.4 % at steady state.

What they do not establish. (i) The growth of higher-k fluctuations by the strong pump, which in our model is the oscillating two-stream instability (we take this to be the process described in the HHS abstract; that identification is our reading, not something we tested against their simulations), has its onset is at $\omega_{pe}t \simeq 783.5$–862 at $v_q/v_{te} = 0.07$, at the end of our window. The windows

end by positivity loss at nearly the same times, because the numerical instability of the Hermite basis reaches the resolved range there, so we cannot say whether the transfer would saturate after the cascade develops. The bound is a statement about the interval before it. HHS state that the deposited energy is limited to about the electron thermal energy [1]; in our runs about 97 % of the transferred energy is still coherent $k = 0$ quiver energy at the window end and the thermal gain is $0.24\, nT_eV$. Whether their "deposited energy" counts the coherent part is not something we can settle from the abstract and the v1 text we read, so we do not claim a contradiction; the claim for times after the cascade is neither confirmed nor refuted here. (ii) The model is nonrelativistic and one-dimensional, with a box $L = 40\, c/\omega_{pe}$ and the strict 2/3 mask retaining Fourier modes $k \leq 9$ at $N_x = 32$ and $k \leq 20$ at $N_x = 64$ (in units of $2\pi/L$), against 1000 cells in HHS; the retained modes cover the onset but not a developed spectrum. (iii) The seed amplitude is imposed; at this seed the deficit is 0.68 %, and a different fluctuation level rescales it. (iv) Drives above $v_q/v_{te} = 0.07$ (including $v_q/v_{te} = 0.1$) have no window beyond about 760. (v) Our claims are bounds for $v_q/v_{te} \leq 0.07$ and $t \lesssim 790$ at the strongest drive. We make no claim beyond.

Why a nonlinear detuning could be expected. The nonlinear trapped-particle timescale of O'Neil [14] is the inverse bounce frequency, $\tau = (m/eEk)^{1/2}$, with linear theory failing after τ; and the ponderomotive coupling between the pump and the plasma is the origin of the modulational and collapse processes of Zakharov [15]. For the uniform pump ($k = 0$ drive) the mean-field law is exact and the trapped timescale is not relevant until finite-k fluctuations grow, which is why the first deviations in our model are the finite-k OTSI.

VIII. CONCLUSIONS

We built and verified a noise-free Hermite–Fourier solver for the canonical Maxwell–Proca–Vlasov system. Its limit is a numerical truncation instability of the asymmetrically weighted Hermite basis, which bounds certified windows to 1000, 892 and 791.5 at $v_q/v_{te} = 0.03$, 0.05, 0.07; no higher resolution, stronger closure, or alternative basis we tried (Dai, symmetric weight, square-root form) removes it. Inside those windows (R1), the energy transfer follows the linear resonant law to 0.68 %, with no resolved frequency detuning, finite dark-reservoir depletion and finite coherence follow the linear predictions, and the small deficit is seed back-reaction. With collisions (R2), the absorption is the linear Drude law with $\nu_{\text{eff}} = (\nu_{ei}/2)(1 + m_e/m_i)$, quadratic in the drive, and independent of the collision form. These are bounds in a restricted model. They do not test nonlinear suppression after the oscillating two-stream cascade develops, relativistic electrons, $k \neq 0$ coupling, or a temperature-dependent collision frequency; a stable basis is needed to extend the windows.

DATA AND CODE AVAILABILITY

All code, records and the generated results are in the Dark-SPECTRAX repository [16] (`https://github.com/uwplasma/Dark-SPECTRAX`); this manuscript was generated against the repository commit recorded in the `paper/` directory history. Figures, tables and every number quoted from a record are produced by `paper/make_figures.py`, which writes `paper/numbers.tex` with the record path of each value as a comment. The parent solver is SPECTRAX [6] at the commits recorded in the JSON files: `35231ab9` (full hash in `pyproject.toml` and the R2 records; the pin in `pyproject.toml`, used for R2), `4090fbe` (R1 nonlinear) and `3de6b2b` (R1); this is the `integration/dark-baseline` branch, which collects open parent PRs, and is not a release. Records: `studies/r1`, `studies/r1nl`, `studies/r2`; gates were committed in `studies/r1_common.py`, `studies/r1nl_gates.py` and `studies/r2_gates.py` before the runs.

ACKNOWLEDGMENTS

[Acknowledgments placeholder.]

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